

Metal speciation and environmental impact on sandy beaches due to El Salvador copper mine, Chile

Marco Ramirez ^a, Serena Massolo ^{b,*}, Roberto Frache ^b, Juan A. Correa ^a

^a *Facultad de Ciencias Biológicas, Departamento de Ecología y Center for Advanced Studies in Ecology and Biodiversity,*

P. Universidad Católica de Chile, Casilla 144-D, Santiago, Chile

^b *Dipartimento di Chimica e Chimica Industriale, Sezione di Chimica Analitica ed Ambientale, Università di Genova, Via Dodecaneso 31, 16146-Genova, Italy*

Abstract

Several coastal rocky shores in northern Chile have been affected by the discharges of copper mine tailings. The present study aims to analyze the chemical speciation of heavy metals in relation to the diversity of sessile species in the rocky intertidal benthic community on the northern Chilean coast, which is influenced by the presence of copper mine tailings.

In particular, the chemical forms of Cd, Cu, Fe, Mn, Ni, Pb and Zn in beach sediment samples collected in the area influenced by El Salvador mine tailings were studied using a sequential chemical extraction method.

In general, all the elements present a maximum concentration in the area near the actual discharge point (Caleta Palito). With regard to Cu and Mn, the concentrations range between 7.2–985 and 746–22,739 µg/g respectively, being lower than background levels only in the control site of Caleta Zenteno. Moreover, the correlation coefficients highlight that Fe, Mn and Ni correlate significantly and positively in the studied area, showing a possible common, natural origin, whilst Cu shows a negative correlation with Fe, Mn and Ni. It could be possible that Cu has an anthropogenic origin, coming from mining activity in the area.

Cd, Fe, Mn, Ni, Pb and Zn are mostly associated with the residual phase, whilst Cu presents a different speciation pattern, as resulted from selective extractions. In fact, Cu is highly associated with organic and exchangeable phases in contaminated localities, whilst it is mainly bound to the residual phase in control sites. Moreover, our results, compared to local biological diversity, showed that those sites characterized by the highest metal concentrations in bioavailable phase had the lowest biodiversity.

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1. Introduction

1.1. Mining history

Copper mining in Chile is based in open or underground mines spread along the Andes Mountains. Porphyry deposits, which are the world's principal source of copper and molybdenum, characterize this area.

Ore minerals (mainly Cu and Mo sulphides) are separated from gangue minerals and pyrite by flotation (Dold and Fontbotè, 2001). Generally, most mining operations, such as processing, smelting and tailing disposal, are carried out near the exploitation areas. However, the El Salvador mine, a porphyry copper deposit located in the Atacama Desert, is an exception because the tailings were dumped without treatment directly into Chañaral Bay via the river Salado (Castilla, 1983; Paskoff and Petiot, 1990). Approximately 150 million tonnes (mining between 1938 and 1975) of disposed materials accumulated in the area have caused a beach to widen (Castilla and Correa, 1997). In 1976 the

* Corresponding author. Tel.: +39 010 3536178; fax: +39 010 3536190.

E-mail address: massolo@chimica.unige.it (S. Massolo).

discharge was diverted via a canal to the rocky beach of Caleta Palito, about 10 km north of Chañaral Bay. Between 1976 and 1989, Caleta Palito received about 130,000 million metric tonnes of mine wastes containing a total copper concentration of 6000–7000 µg/l, thereby extending La Lancha in the northern area. Approximately 70% of tailing sediments were trapped by the Bay whereas 30% left the Bay, thus at least 9 m tailing sediments remained deposited at the centre of the artificial beach (Castilla, 1983).

In 1990 an environmental court action ruled that a settlement dam should be constructed in the desert between the El Salvador mine and the coast and that only “clear water” tailings, containing no more than 2000 µg/l total of copper should be dumped at Caleta Palito (Lee et al., 2001).

In 1995, a study on the coastal ecosystem around Caleta Palito was carried out to understand the effects of Cu on the local flora and fauna.

1.2. Metal distribution and bioavailability

The most important effects of the disposal of untreated tailings include an increased copper concentration in the water at the impacted beaches, widened beaches and the elimination of invertebrates and algae around the dumping sites. Reduced biodiversity and destruction of the trophic chains together with a lower coverage of species in rocky intertidal communities are the observed ecological effects (Castilla, 1983, 1996; Correa et al., 1999, 2000; Fariña, 2000; Fariña and Castilla, 2001; Lee et al., 2001).

Heavy metal distribution and bioavailability in both sediments and the water column have to be considered to obtain a better understanding of environment–organism interactions. Besides physical–chemical parameters, mining effluent components (in particular, heavy metals and sediments) are the most important factors directly and indirectly influencing the coastal marine community structure (Ellis, 1987; Fariña and Castilla, 2001).

Sediments are the final destination of trace metals, as a result of adsorption, desorption, precipitation, diffusion processes, chemical reactions, biological activity and a combination of those phenomena. Sediments are an important sink for heavy metals but when some physical disturbance occurs, or there is diagenesis and/or changes in pH or redox potential, they can become a source of metals, releasing them in the overlying water column. This phenomenon can occur even long after the end of direct discharge and its extent depends on the metal association with the different mineralogical fractions of the sediment, defined as “solid speciation”. Therefore, metal behaviour and availability strictly depends upon their chemical form and therefore their speciation (Jones and Turki, 1997).

Total metal concentration is not sufficient to assess the environmental impact of polluted sediments since heavy metals may have different chemical forms and only a fraction can be remobilized easily.

Studies on the distribution and speciation of heavy metals in sediments can provide not only information on the degree of pollution, but especially the actual environmental impact on metal bioavailability as well as their origin.

To date, it has generally been accepted that the most appropriate methods to evaluate solid speciation—defined as the identification and quantification of the different species, forms or phases present in sediment—are selective sequential extraction procedures (Kot and Namiesnik, 2000). Selective extractions are widely used in sediment analysis to evaluate long-term potential emission of pollutants and to study the distribution of pollutants among the geochemical phases (Rauret, 1998), and to determine the metals associated with source constituents in sedimentary deposits (Van der Sloot et al., 1997). According to Rubio et al. (1991), metals with an anthropogenic origin are mainly extracted in the first step of the procedure, while lithogenic metals are found in the last step of the process corresponding to the residual fraction.

This study aims to evaluate the fate of suspended sediments from the El Salvador mine and to provide information on enrichment and speciation of some heavy metals (Cd, Cu, Fe, Mn, Ni, Pb and Zn) in sediments from the area influenced by El Salvador mine tailings. The results are discussed in relation to the geological characteristics to assess the extent of anthropogenic input in the investigated area. As previously mentioned, although several studies on the coastal area around Caleta Palito have been carried out, the role of sediments in environment–organism interactions was not considered.

2. Material and methods

2.1. Sample collection and pre-treatment

Sediment samples were collected in summer 2002 at 16 stations located at various distances from the discharge point at Caleta Palito (26°15'S; 69°34'W) covering about 90 km of coastline.

Fig. 1 shows the map of the area with the position of the sampling sites, which were divided “a priori” into two groups on the basis of results obtained in previous studies (Lee et al., 2002; Correa et al., 1999): reference sites (Pan de Azúcar Norte, Pan de Azúcar Sur and Caleta Zenteno) and impacted sites.

Sediment samples were collected in sandy beaches using a plastic spoon washed with 10% nitric acid and rinsed with Milli-Q water to avoid any contamination. The samples were put in polyethylene bags and stored

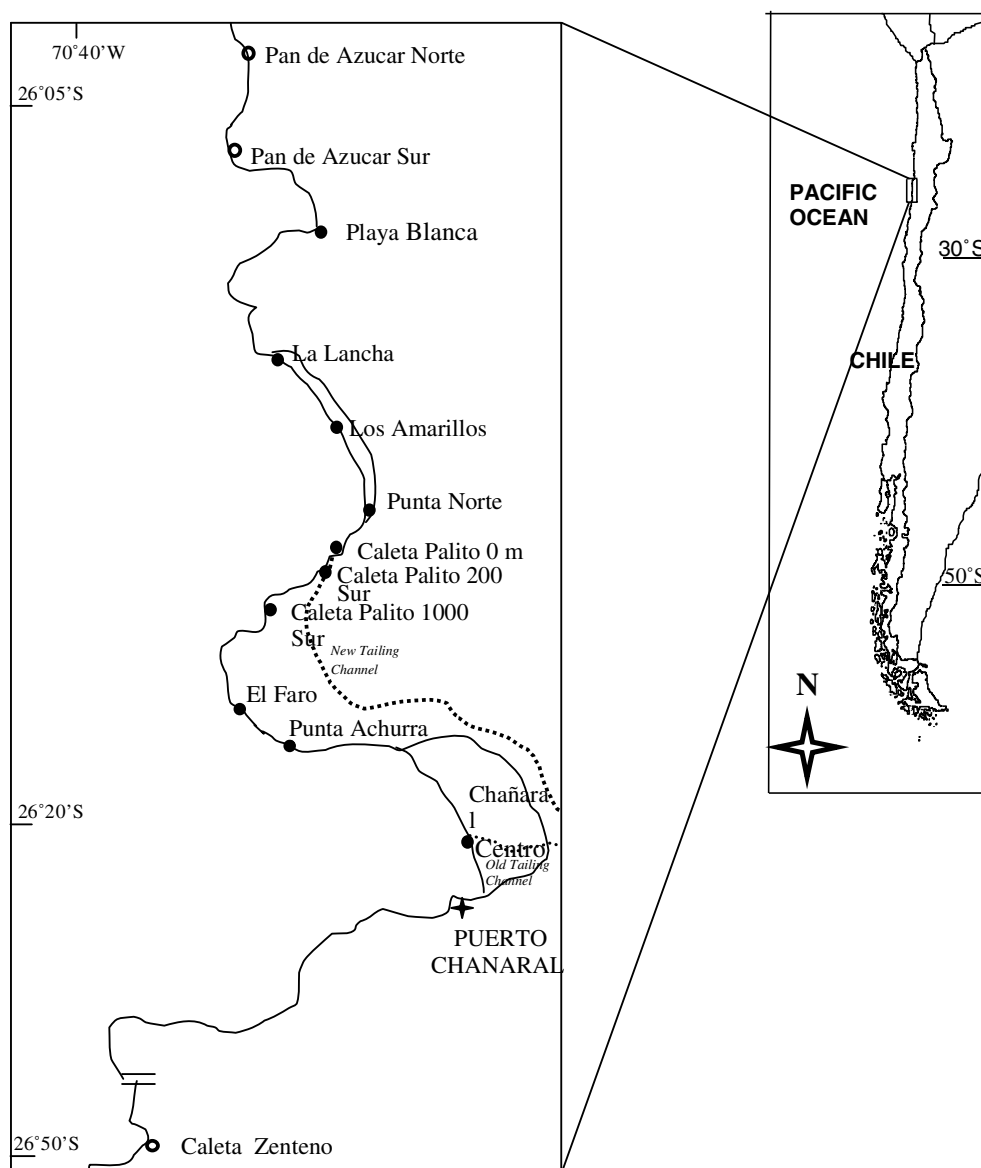


Fig. 1. Map of the studied area and sampling stations location (black dots stand for impacted sites, whilst white dots for control sites).

in dark, cold conditions (+4°C). The samples were sieved in laboratory: the fraction exceeding 1.25 mm was broken up and not analyzed whilst the remainder sediments were used for metal determination. When there was enough fine material, fraction <1.25 mm was separated into two size fractions with a sieve of 63 µm mesh size to obtain fine fraction (<63 µm) and coarse fraction (63 µm–1.25 mm). All the samples were oven dried at 60°C, homogenized with an automatic agate grinder and stored at room temperature until analysis.

2.2. Pseudototal attack

Sediment samples were digested in PTFE vessels with aqua regia (HCl:HNO₃ 3:1) in a 650 W microwave oven (CEM MDS 2000) with the following program: 5 min at

40% power, 5 min at 60% power and 10 min at 80% power. The digested samples were filtered, transferred to polyethylene containers and stored at +4°C until analysis. Reagent blank was processed with the samples and it did not show any significant contamination. Accuracy of the procedure was checked using CRM MESS 2 marine sediment certified by the National Research Council of Canada for the metal content.

2.3. Selective extraction

Selective extraction is based on the procedure used by Tessier et al. (1979), already modified in recent years (Baffi et al., 1998), with improvements made according to the European Community Bureau of Reference (BCR 701), which examined and finally eliminated irre-

producibility sources. It is made up of three steps, which dissolve the following phases respectively: exchangeable and bound to carbonate, bound to Fe and Mn oxides and hydroxides, bound to organic matter and sulphides.

Exchangeable and bound to carbonate phase (phase 1) is extracted with 0.11 M acetic acid, while the fraction bound to Fe–Mn oxides (phase 2) with 0.5 M hydroxylamine hydrochloride, adjusted to pH 2 with nitric acid (65%). The phase bound to organic and sulphides (phase 3) is extracted with 8.8 M hydrogen peroxide, treated at 80 °C in a microwave oven using the following program: 10 min at 10% power, 10 min at 0% power, 20 min at 20% power, 10 min at 0% power and 20 min at 30% power, and 2 M ammonium acetate adjusted to pH 2 with nitric acid (65%). Each extraction was carried out overnight (16 h) at room temperature. All the reagents employed were Tracepur grade (Merck Eurolab, Italy).

After each extraction, the samples were separated from the aqueous phase by centrifuging at 4000 rpm for 20 min. The sediments were washed with Milli-Q water and centrifuged again. The wash water was added to supernatants.

The metal content of the residual phase was obtained from the difference between the total content and the sum of phases 1, 2 and 3, according to Ianni et al. (2000, 2001) and Mester et al. (1998). Sequential extraction reagent blanks showed no detectable contamination. Accuracy of the procedure was checked with CRM 701 (SM&T). The recovery rates for heavy metals in the standard reference material ranged between 82% and 110%.

2.4. Metal analysis

The Cu, Fe, Mn, Ni, Pb and Zn concentrations were determined with an inductively coupled plasma atomic emission spectrometer (ICP-OES) Vista Pro (Varian), with the external standard method, using matrix-matching calibrants. Cd was determined by electrothermal atomization atomic absorption spectrometry (ETA-AAS). A Varian Spectra A300 spectrometer with Zeeman effect background correction and autosampler Varian Model 96 was used employing the standard addition method for calibration.

3. Results and discussion

3.1. Comparison with global data

Table 1 reports the mean metal concentration found in non-contaminated sediments used as references for non-contaminated areas (Salomons and Förstner, 1984) and metal ranges found in our study in the area influenced by El Salvador mine, Chañaral Bay.

Table 1

Heavy metal concentrations in non-contaminated sediments (Salomons and Förstner, 1984) in comparison with ranges found in Chañaral area

	Background ($\mu\text{g g}^{-1}$)	Chañaral area ($\mu\text{g g}^{-1}$)
Cd	0.17	0.061–1.085
Cu	33	7.20–1985
Fe	41,000	9055–32,999
Mn	770	746–22,739
Ni	52	0.167–7.57
Pb	19	1.57–21.2
Zn	95	19.8–236

Fe, Ni and Pb content falls below mean values reported for non-contaminated sediments whilst Cd concentrations are higher than the background values in Punta Norte, Caleta Palito 200 Sur, El Faro and Caleta Zenteno, ranging between 0.061 and 1.085 $\mu\text{g/g}$, as can be seen in Table 3. Total Cu and Mn concentrations fluctuate between 7.20–985 and 746–22,739 $\mu\text{g/g}$ respectively, being lower than background values only in the control site of Caleta Zenteno.

Zn concentration in coastal sediments near the discharge point shows values between 19.8 and 236 $\mu\text{g/g}$ with the highest in Punta Norte, Caleta Palito 0 and El Faro.

Apart from Caleta Zenteno, all the studied sites show Cu and Mn contamination, whilst only some localities show Zn and Cd concentrations above background values.

To estimate the possible environmental consequences of the metal analyzed, our results were also compared to US NOAA's sediment quality guidelines. In this study the effects range-low (ERL) and effects range-median (ERM) concentrations are considered. The ERL represents chemical concentrations below which adverse biological effects were rarely observed, while the ERM represents concentrations above which effects were more frequently observed. Generally, adverse effects occurred in less than 10% of studies in which concentrations were below the respective ERL values, and were observed in more than 75% of studies in which concentrations exceeded ERM values (Long et al., 1995, 1997).

ERL and ERM values for the metals object of this study are reported in Table 2.

Table 2

US NOAA's ERL and ERM concentrations for the studied metals (values are in $\mu\text{g g}^{-1}$ dry weight)

	ERL ($\mu\text{g g}^{-1}\text{-dw}$)	ERM ($\mu\text{g g}^{-1}\text{-dw}$)
Cd	1.2	9.6
Cu	34	270
Fe	No values given	No values given
Mn	No values given	No values given
Ni	20.9	51.6
Pb	46.7	218
Zn	150	410

Table 3

Heavy metal total contents ($\mu\text{g g}^{-1}$ dry weight) in sediment samples (data represent the mean $\pm$ standard deviation of 10 determinations)

Samples	Grain size	Cd	Cu	Fe	Mn	Ni	Pb	Zn
Pan de Azucar Norte	<1.25mm	0.061 $\pm$ 0.002	60.4 $\pm$ 0.6	9927 $\pm$ 2	5608 $\pm$ 1	0.57 $\pm$ 0.01	3.67 $\pm$ 0.04	22.6 $\pm$ 0.0
Pan de Azucar Sur	<1.25mm	0.106 $\pm$ 0.002	173 $\pm$ 1	21,873 $\pm$ 2	12567 $\pm$ 2	3.26 $\pm$ 0.01	6.76 $\pm$ 0.12	38.0 $\pm$ 0.1
Playa Blanca	<1.25mm	0.109 $\pm$ 0.002	1736 $\pm$ 1	13252 $\pm$ 2	1709 $\pm$ 3	2.32 $\pm$ 0.02	9.58 $\pm$ 0.03	100 $\pm$ 1
La Lancha	<1.25mm	0.053 $\pm$ 0.001	1831 $\pm$ 1	14373 $\pm$ 3	2065 $\pm$ 3	1.07 $\pm$ 0.03	10.6 $\pm$ 0.1	78.8 $\pm$ 1.0
Los Amarillos	<1.25mm	0.042 $\pm$ 0.001	1985 $\pm$ 1	9055 $\pm$ 2	1240 $\pm$ 2	3.76 $\pm$ 0.01	9.37 $\pm$ 0.10	41.9 $\pm$ 1.0
Punta Norte	<1.25mm	0.194 $\pm$ 0.015	924 $\pm$ 1	32999 $\pm$ 1	22,739 $\pm$ 1	7.57 $\pm$ 0.04	12.7 $\pm$ 0.2	154 $\pm$ 1
Caleta Palito 0	<1.25mm	0.179 $\pm$ 0.015	819 $\pm$ 1	30746 $\pm$ 1	22475 $\pm$ 1	6.16 $\pm$ 0.04	11.7 $\pm$ 0.6	130 $\pm$ 1
Caleta Palito 200 Sur	<1.25mm	0.502 $\pm$ 0.030	569 $\pm$ 1	22,739 $\pm$ 1	16644 $\pm$ 6	4.75 $\pm$ 0.02	6.55 $\pm$ 0.42	44.7 $\pm$ 0.1
Caleta Palito 1000 Sur	<1.25mm	0.152 $\pm$ 0.023	1758 $\pm$ 1	19066 $\pm$ 2	11331 $\pm$ 3	2.67 $\pm$ 0.03	5.47 $\pm$ 0.01	40.0 $\pm$ 0.1
El Faro	<1.25mm	0.225 $\pm$ 0.030	807 $\pm$ 1	20411 $\pm$ 2	6730 $\pm$ 2	2.86 $\pm$ 0.01	6.96 $\pm$ 0.21	236 $\pm$ 1
Chañaral Centro	<1.25mm	0.093 $\pm$ 0.010	1659 $\pm$ 1	12541 $\pm$ 3	2367 $\pm$ 2	0.17 $\pm$ 0.01	21.2 $\pm$ 0.1	28.1 $\pm$ 2.9
Caleta Zenteno	<1.25mm	0.477 $\pm$ 0.023	7.20 $\pm$ 0.02	15966 $\pm$ 2	746 $\pm$ 3	0.27 $\pm$ 0.02	1.57 $\pm$ 0.10	24.5 $\pm$ 0.2
El Faro	<63 μm	0.802 $\pm$ 0.057	1896 $\pm$ 1	22610 $\pm$ 2	5466 $\pm$ 4	13.6 $\pm$ 0.1	15.6 $\pm$ 0.2	259 $\pm$ 1
El Faro	63 μm –1.25mm	0.169 $\pm$ 0.015	689 $\pm$ 1	19357 $\pm$ 3	8524 $\pm$ 1	5.96 $\pm$ 0.01	5.06 $\pm$ 0.34	223 $\pm$ 1
Punta Achurra	<63 μm	0.896 $\pm$ 0.055	2116 $\pm$ 1	35891 $\pm$ 2	4001 $\pm$ 2	5.67 $\pm$ 0.03	18.5 $\pm$ 0.5	519 $\pm$ 2
Punta Achurra	63 μm –1.25mm	0.118 $\pm$ 0.015	687 $\pm$ 1	22191 $\pm$ 4	4180 $\pm$ 4	1.17 $\pm$ 0.01	6.37 $\pm$ 0.04	331 $\pm$ 2
Chañaral Centro	<63 μm	1.09 $\pm$ 0.02	1259 $\pm$ 1	17307 $\pm$ 4	2825 $\pm$ 4	7.77 $\pm$ 0.02	9.47 $\pm$ 0.07	55.3 $\pm$ 3.2
Chañaral Centro	63 μm –1.25mm	0.173 $\pm$ 0.057	756 $\pm$ 1	14713 $\pm$ 4	2855 $\pm$ 3	0.17 $\pm$ 0.02	10.7 $\pm$ 0.2	19.8 $\pm$ 2.3

Comparing our data with ERL and ERM values, all the metals, apart from Cu, show lower concentrations than ERL. In the case of Cu, though, all the studied sites, except for Caleta Zenteno, show higher concentrations than the ERM value. In particular, Cu concentration is almost five times higher than the ERM value for all the contaminated sites. Considering that toxicity is a function also of the degree to which data exceed ERM values, we can expect some environmental or toxicological effect of this metal.

The total heavy metal concentration in sediments is reported in Table 3.

In general, the samples collected north of the actual discharge point (Caleta Palito 0) have the highest concentrations of all the elements. Moving away from this area, the heavy metal levels progressively decrease, reaching very low values in control sites (Caleta Zenteno in the south and Pan de Azucar Norte in the north). Thus, this feature confirms the effect of the tailing discharge.

Exceptions to this general pattern are represented by Cd and Pb, which reach maximum values in Caleta Zenteno and in Chañaral Centro respectively. The maximum Pb value found in Chañaral may be due to anthropic activities, as Chañaral the only town present in the studied area.

The increase in metal concentration and the formation of a new beach in the area north of the discharge point suggests that the main long shore current is directed northwards.

Only for samples taken from El Faro, Punta Achurra and Chañaral Centro was it possible to separate fractions <63 μm (fine fraction) and >63 μm (coarse fraction). Apart from Mn, the concentrations of all metals are much higher in fine than in coarse fraction. This pat-

tern is particularly evident for Cd, whose concentrations are only over background value in fraction <63 μm because its specific surface, that is larger than in the coarse fraction, facilitates absorption processes. Nevertheless, fine fraction represents less than 10% in these areas so that its contribution over the whole sediment is not significant. Cu concentrations in both fine and coarse fractions are compared in Fig. 2.

The three sediment samples for which it was possible to separate two grain size fractions were collected south of the actual discharge point, suggesting that the fine fraction is probably composed of tailing sediments coming from the El Salvador mine and discharged onto the sandy beach of Chañaral bay up to 1975. The absence of fine fraction (<63 μm) in sediments sampled north of Caleta Palito might be explained by differences in treatment and elimination procedures in the two historical dumping sites (i.e. Chañaral Bay and Caleta Palito). In fact, starting from 1990 an environmental court

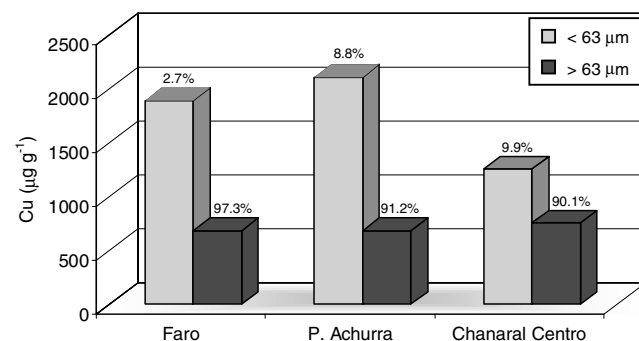


Fig. 2. Cu total contents in different grain size <63 μm and between 63 μm and 1.25mm. Numbers above the histogram bars refer to relative weight percentage of each granulometric fraction.

action ruled that only “clear water” tailings could be discharged into the sea.

The correlation coefficient matrix ($p = 1\%$) among the heavy metals contents is reported in Table 4.

As can be seen, Fe, Mn and Ni correlate significantly and positively in the studied area, showing a possible

common, natural origin (Rivaro et al., 1998), whilst Cu has a negative correlation with Fe, Mn and Ni. Cu could have an anthropogenic origin, coming from mining activity in the area, as confirmed by the total Cu concentration, which is higher than the background value. Cd, Pb and Zn do not show any correlation with other studied metals.

Table 4

Correlation matrix ($p = 1\%$) among total metal concentrations in sediments

	Cd	Cu	Zn	Pb	Ni	Mn	Fe
Cd							
Cu	−0.43		0.10	0.55	−0.09	−0.13	−0.15
Zn	0.26	0.10		0.31	0.38	0.19	0.40
Pb	−0.29	0.55	0.31		0.40	0.37	0.36
Ni	0.02	−0.09	0.38	0.40		0.97	0.97
Mn	−0.07	−0.13	0.19	0.37	0.97		0.93
Fe	0.12	−0.15	0.40	0.36	0.97	0.93	

3.2. Metal speciation results

Fig. 3 reports histograms representing the results of selective extractions.

The highest Cd, Fe, Mn, Ni, Pb and Zn concentrations are found in the residual phase. In particular, Cd, Fe and Mn residual phase content represents more than 90% of the total. As regards Fe, phase 2 (Fe and Mn oxides and hydroxides) represents about 10% of the total amount. Fe speciation shows that ferric

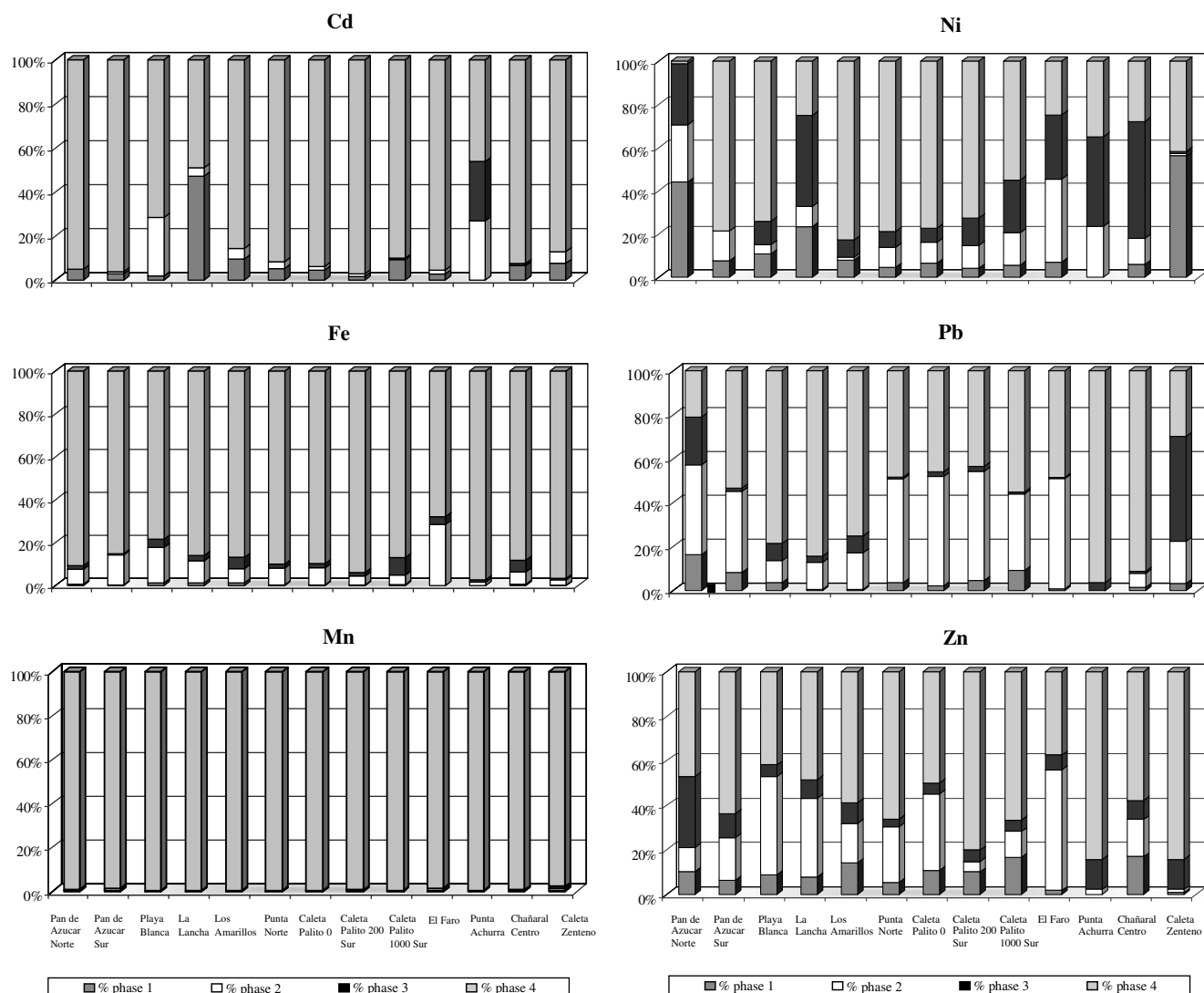


Fig. 3. Results of selective extraction for Cd, Fe, Mn, Ni, Pb and Zn in the sediments.

oxyhydroxide content is low in relation to the relatively high pyrite content and this sparseness is due partly to Mo poisoning of sulphide oxidizing bacteria (Dold and Fontbotè, 2001).

Cd shows some differences among the samples: for example the residual phase is lower than 60% in La Lancha and Punta Achurra, while it reaches 90% in the other samples.

The percentage of Ni, Pb and Zn in the residual phase is lower than the percentage of Fe, Mn and Cd and the samples show differences with regard to the speciation of these metals. More than 40% of Ni is present in the exchangeable phase in control sites, such as Caleta Zenteno and Pan de Azucar Norte. There is about 30% of Ni associated to phase 3 in Playa Blanca and in areas between Caleta Palito and Chañaral Centro, while Ni associated to the residual phase ranges from 30% to 80% of total concentration. The sediments sampled in control areas exhibit a different speciation from the other sites as regards Pb. In Pan de Azucar Norte and in Caleta Zenteno about 25% of total concentration is associated to the residual phase and about 30% to organic matter and sulphides. In the other sediments Pb is associated to the residual phase for more than 50% of total concentration and a large percentage of Pb is also associated to the Fe–Mn oxides phase. More than 50% of the total concentration of Zn is present in the residual phase, whilst there is 10–30% in the reducible phase.

Cd, Mn and Zn concentrations measured in phase 1 are very low, limiting their potential toxicity as pollutants, despite the total concentrations for these metals being higher than the background values.

With respect to the other metals studied, Cu presents a different speciation with a low percentage of total concentration in the residual phase, as reported in Fig. 4.

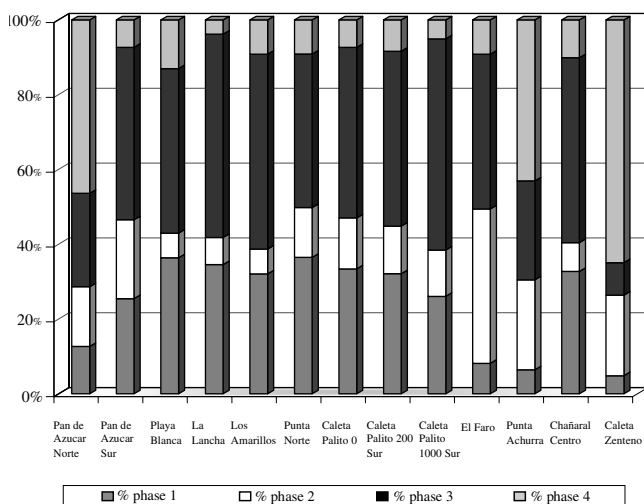


Fig. 4. Results of selective extraction for Cu in unsieved sediments.

In the area affected by mine tailings Cu is bound to residual phase for about 10% of total concentration, to oxidable phase for 40% and to labile phase for 30%; the Cu in residual phase constitutes more than 50% of total concentration only in two control sites (Pan de Azucar Norte and Caleta Zenteno). This confirms the high affinity of Cu to organic matter, and it could in fact easily form complexes with organic matter due to the high stability constant of organic-Cu compounds (Xiangdong et al., 2001).

Cu concentrations found in sediments in the four geochemical phases are shown in detail in Fig. 5.

Each phase shows very low Cu concentrations in control sites (Pan de Azucar Norte, Pan de Azucar Sur and Caleta Zenteno) in comparison to the other studied areas, despite the total amount of Cu being lower than background levels only in Caleta Zenteno.

As already observed, Cu speciation is very similar for all the samples apart from Pan de Azucar Norte and Caleta Zenteno, where the residual phase is prevalent. However, the other sites can be divided into two groups according to their concentration ranges. Cu concentrations are lower in Punta Norte, Caleta Palito 0, Caleta Palito 200 Sur, El Faro and Punta Achurra than in Playa Blanca, La Lancha, Los Amarillos, Caleta Palito 1000 Sur and Chañaral Centro samples. In the former group Cu concentration fluctuates between 50 and 340 µg/g in phase 1 and between 200 and 400 µg/g in phase 2, whilst in the latter it ranges from 460 to 640 µg/g and from 750 to 1000 µg/g, respectively.

It is evident that the highest Cu values are found in small bays located north of the actual discharge point. In this context, local hydrodynamics may play an important role, transporting contaminated sediments from the discharge point northwards to other beaches. As previously noted by other authors (Castilla, 1983), this region is characterized by high water dynamics resulting in sediments closely related to copper tailings being transported to the beaches without any alteration, as proved by mineralogical studies of sediment samples. On the other hand, high Cu levels found in Chañaral Centro and in Caleta Palito 1000 Sur may be related to the effects of the first discharge site.

Coastline topography also plays a significant role in Cu accumulation processes. In those sites protected by a physical barrier, for example promontory, the transport and subsequent the deposition of sediments is impeded, thereby reducing Cu contamination. In particular, El Faro, Punta Achurra and Punta Norte sediments show lower Cu concentration than the other sites despite being close to the discharge point. Moreover, these samples show lower Cu concentration in exchangeable phase, confirming that Cu input is not recent. On the other hand, high metal concentration in labile phase could be related to recent coastal input.

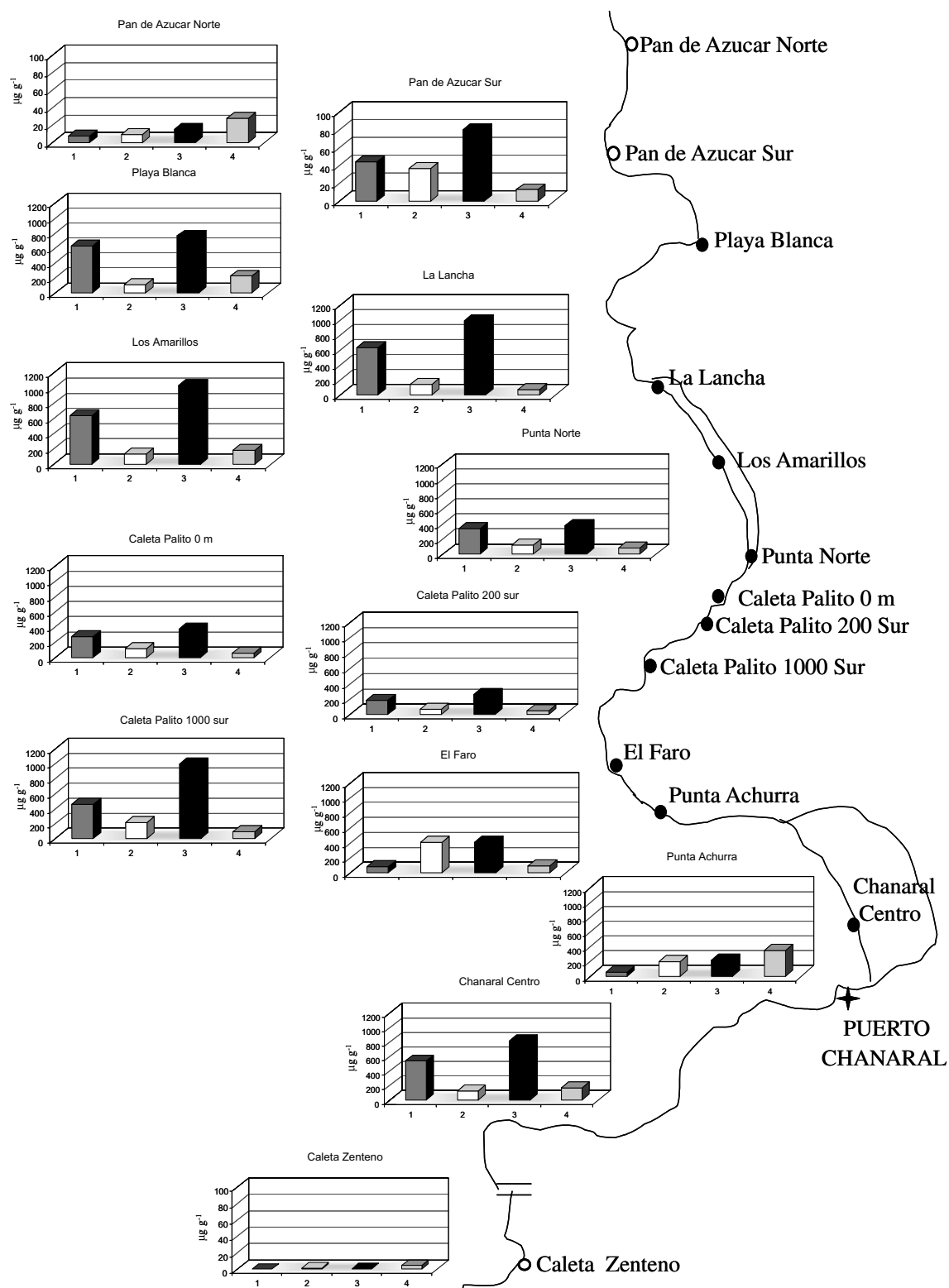


Fig. 5. Cu concentrations in phase 1 (exchangeable and bound to carbonates), 2 (bound to Fe and Mn oxides), 3 (bound to organic matter and sulphides) and 4 (residual) in the studied area.

Cu speciation in the two size fractions is compared in Fig. 6.

El Faro and Chañaral fine and coarse fractions present the same speciation pattern, which is similar to that

the one at Chañaral (Correa et al., 1999; Lee et al., 2002). Moreover, Correa et al. (2000) rejected the hypothesis that Cu alone, at concentration occurring in seawater in the vicinity of the discharge point in Caleta Palito, is responsible for the low algal diversity.

Cu concentration data found in sediments agree with biological diversity: in fact, Cu concentration at La Lancha is twice the amount found at Caleta Palito both as total concentration and labile phase. These observations highlight a possible role of sediments in regulating the biological population. From a biological point of view, the lower diversity and density found in these sites is directly correlated with a concentration increase in the exchangeable and bound to organic matter and sulphides phase.

From 1990 only clear water tailings and not solids were dumped at Caleta Palito, resulting in significant differences in the biological community. Biological studies (Correa et al., 2000) suggest that despite the mine's negative impact in the past on the algal assemblages in the impacted beaches, today's situation regarding algal diversity and abundance seems to depend on a further factor: the large abundance of herbivores without any predators regulating their population size. Toxicity studies based on the algal copper tolerance excluded, in fact, that copper is responsible for preventing the algal growth.

The meiofauna, unlike algae and macrofauna, spend their entire life cycle within the sedimentary environment and is consequently more responsive to the input of a pollutant to the sedimentary environment than the macrofauna (Coull and Chandler, 1992). The impacted beaches are characterized by the absence or near absence of copepods, suggesting that they are useful as indicators of pollution stress (Lee et al., 2001). These studies highlight the importance of heavy metal associated at the sediment for biological population. Lee et al. (2001) determined that metal enrichment generally drives down both diversity and density of meiofaunal assemblages.

4. Conclusions

In this study, we analyzed heavy metal distribution and speciation in sediments collected in the coastal areas of El Salvador mine (Chile).

The correlation of the concentration with the sediment grain size confirmed preferential association of metals with fine fraction.

Comparing total concentrations in the sediments with those reported for non-contaminated sediments, it is evident that for Cu, Mn, Zn and Cd there is some enrichment.

The metal speciation analysis provided information on their bioavailability and mobility, which is easier

for those metals bound to labile phases and showed that the metals depended on their origin.

Most of Cd, Mn and Zn (even if with a lower percentage) are not immediately bioavailable, being present in the refractory phases of the sediments. On the other hand, Cu in the area affected by El Salvador mine tailings is prevalently of recent origin and rather bioavailable. This suggests that Cu has no lithogenic origin but it seems to be associated with mine tailings. For years it was transported both as solid (earlier than 1990) and dissolved form to the sea. Our results demonstrated that Cu in dissolved form could easily be adsorbed to sediments.

The results obtained from the sediment speciation analysis in the Chañaral and Caleta Palito areas enable us to explain the pattern of variance in diversity: sites with the highest metal concentrations in phases 1 and 3 show the lowest diversity. Therefore it may be asserted that studies on metal speciation in sediments can be useful means to understand the responses of biological communities. The evidences presented in this work support the toxicity of Cu when present in large concentration not only in seawater or porewater but also in sediments. Species living in close contact with sedimentary environment show that their density population and abundance fall where bioavailable metal concentrations in sediments are high.

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